

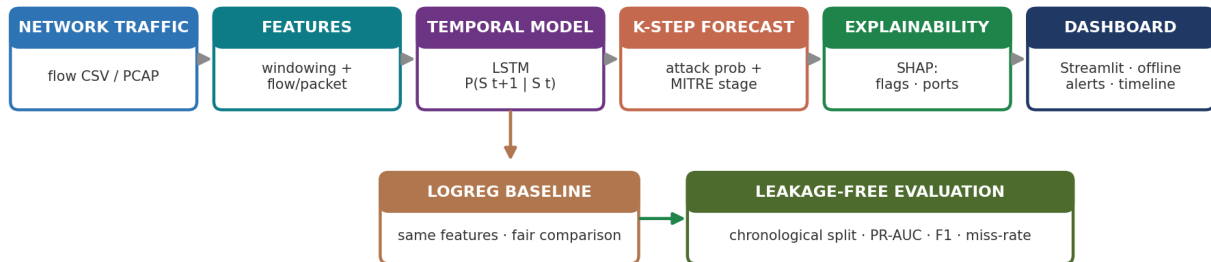
ThreatCast — System Architecture Document

AI-based Network Attack Forecasting · SIH26153 · Team Foresight · Smart India Hackathon 2026

1. Overview

ThreatCast forecasts network attacks a few time-windows **before** they fully unfold, instead of only detecting them once in progress. It reads network traffic, groups it into short time-windows, and a temporal LSTM predicts the attack probability and likely MITRE ATT&CK stage for the next windows, with a SHAP reason for each alert. It runs fully offline on open-source tools and free public data, and is measured honestly against a logistic-regression baseline.

2. System architecture



Data flow: raw traffic → windowing + feature extraction (flow and packet level) → LSTM reads the recent window sequence → K-step forecast (attack probability + MITRE stage) → SHAP explanation → offline dashboard. A logistic-regression baseline runs on the same features for a fair comparison, and all metrics come from a leakage-free evaluation.

3. Components

- **Ingestion & windowing** — reads flow CSVs (and PCAP via Scapy/PyShark) and groups records into fixed time-windows (default 10s).
- **Feature extraction** — flow-level features (ports, TCP flags, bytes, packets, timing) today; packet-level features (TTL, port-scan, retransmissions) on the roadmap.
- **Temporal model (LSTM)** — reads the last H windows (default 8) and forecasts the next K windows (default 3 = 30s ahead).
- **Forecast head** — outputs attack probability and the likely MITRE ATT&CK stage.
- **Explainability (SHAP)** — ranks the driving features per flagged window (e.g. uniq_dport, flow_count).
- **Offline dashboard (Streamlit)** — probability timeline, top-risk windows, and per-alert reasons — no cloud.

4. Evaluation methodology

Evaluation is **leakage-free**: the model trains on earlier traffic and is tested on later traffic (chronological split), and the alert threshold is chosen on a validation split only — so the model cannot peek into the future. Measured on CIC-IDS-2018 (Infiltration day).

Metric	Baseline (LogReg)	ThreatCast (LSTM)
PR-AUC @ 30s ahead	0.76	0.89 (+0.13)
F1 (best threshold)	—	0.69
Horizon held	—	up to 120s (PR-AUC ~0.87)

Honest limitation: at a usable operating point (~0.5% false-alarm target) the current model misses ~47% of attacks. We report the real number and are reducing it (see roadmap).

5. Technology stack

Layer	Tool	Why
Traffic ingestion	Scapy / PyShark	read flow CSV + PCAP
Temporal model	PyTorch LSTM	learn the time pattern $P(S_{t+1} S_t)$
Baseline	scikit-learn (LogReg)	fair, same-feature comparison
Explainability	SHAP	reason for every alert (no black box)
Dashboard	Streamlit	offline SOC view, no cloud
Framework	MITRE ATT&CK	standard attack-stage taxonomy

6. Deployment & compliance

- **Offline / air-gapped** — trains and runs on a normal laptop (CPU); no cloud, suitable for sensitive government networks.
- **Add-on, not a replacement** — acts as an early-warning layer alongside existing IDS/NDR.
- **Sponsor mandate** — protecting Critical Information Infrastructure is NCIIPC's remit; NCIIPC is a unit of NTRO (the problem sponsor), IT Act §70A. Early warning + evidence also support CERT-In (§70B) reporting.

7. Roadmap

- **Lead-time metric** — median seconds of early warning + recall-vs-lead-time curve (highest priority).
- **Packet-level features** — TTL, port-scan, retransmissions (named in the PS); also lowers miss rate.
- **Concrete MITRE stage mapping** — map dataset labels to the 5 stages for a real stage output.
- **Second-dataset validation** — another attack day / CTU-13 to show it generalizes.
- **Packaging** — demo video, cleaner repo, and this document kept current.

Repository: github.com/GMinnu/SIH26153_AI-based-Network-Attack-Forecasting-from-Network-Traffic-Data